Smoker Status Prediction

Using Clinical Biomarkers & Ensemble Machine Learning

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Data Mining · UNC Chapel Hill
github.com/AidanColvin/ml-smoker-status-prediction

Generative AI provided limited assistance in line with the course policy on usage of AI.

0.8887	15K	24
Test AUC	Training Obs	Biomarkers

Agenda

01 **Clinical Problem**
Why self-report fails — 20–30% of smokers deny tobacco use

02 **Data & Dataset Quirks**
15K synthetic Korean health records — provenance and limitations

03 **Exploratory Analysis**
Hemoglobin biology, feature correlations, class distribution

04 **The Research: Feature Biology**
Why each biomarker associates with smoking — published evidence

05 **Methodology**
LassoCV selection · Logistic Regression · Random Forest · Gradient Boosting

06 **Results & Interpretation**
AUC 0.8887 · Feature importance · Calibration · Threshold analysis

07 **Conclusions & Next Steps**
Path to 0.90+ AUC and clinical deployment

01 · Clinical Problem: Why Self-Report Fails

Smoking is the #1 preventable cause of death globally.

Self-reported smoking status is unreliable.

Published estimates place the denial rate among current smokers at **20–30%** in clinical settings. Social desirability bias and fear of treatment consequences drive this underreporting.

If routine laboratory measurements can predict smoking status without self-report, clinicians can proactively identify at-risk patients and intervene.

Can 24 routine biomarkers replace self-report for smoking classification?

1 in 4

Smokers deny use in clinical settings

AUC 0.8887

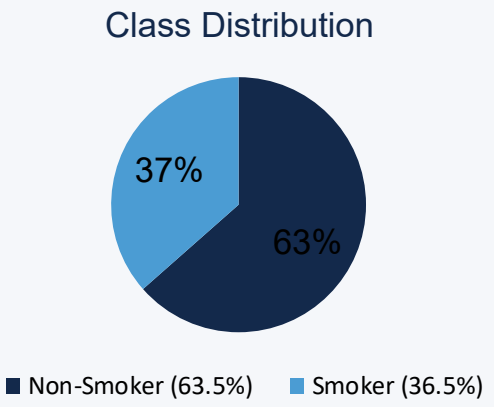
Achieved without any self-report

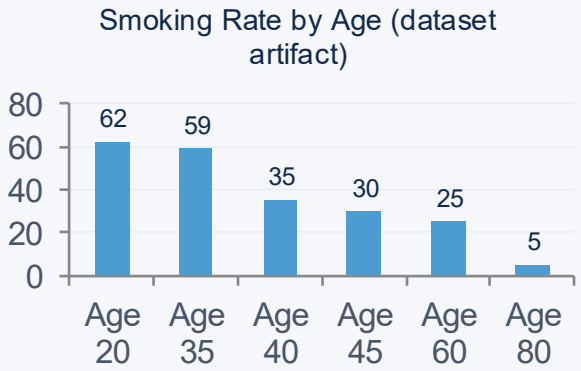
21.2% FNR

False negative rate — Gradient Boosting

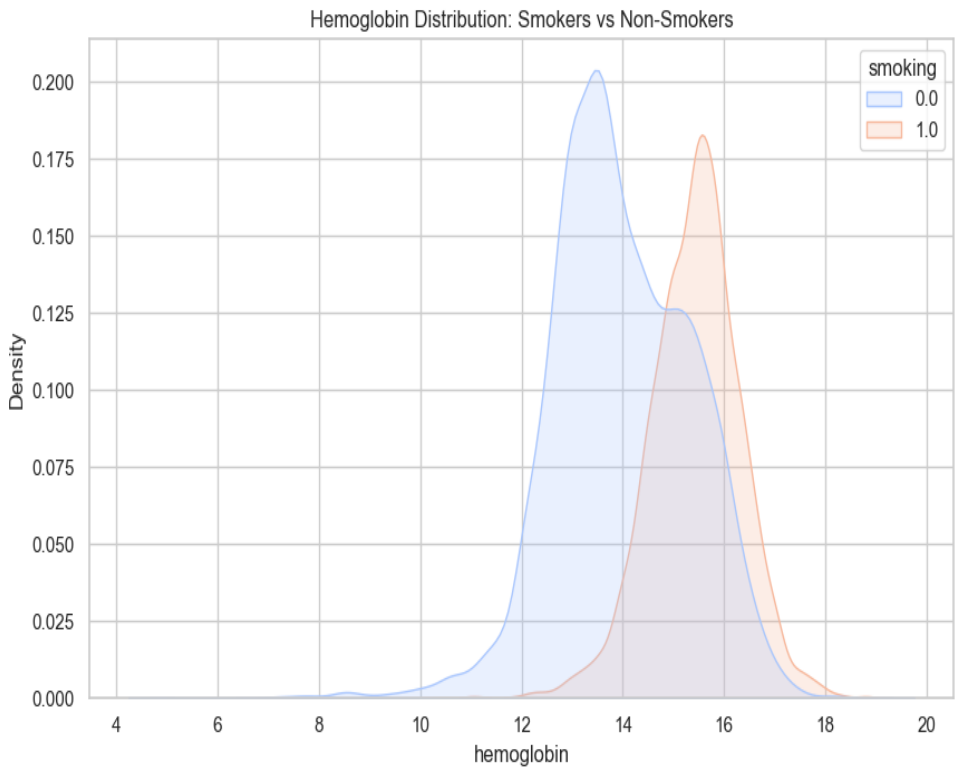
02 · Dataset Overview: 15K Synthetic Korean Health Records

Property	Value	Implication
Source population	Korean gov. health screening	Not generalisable to Western populations
Generation method	Deep-learning generative model	Tail compression in skewed features
Training observations	15,000	~10× smaller than original; limits ceiling AUC
Original dataset size	159,256 rows	Full data = most direct path to 0.90+
Age 40 concentration	35.9% of all observations	Model undertrained at ages <30 or >60
Class imbalance	63.5% non-smoker / 36.5% smoker	Stratified CV sufficient; no SMOTE needed
Sex field	Absent	Height absorbs 41pp sex-by-smoking confound
Alcohol variable	Absent	Gtp/ALT/AST cannot separate tobacco vs. alcohol





03 · EDA: Hemoglobin: The Dominant Signal



~13.8 g/dL

Non-Smoker Mode

~15.7 g/dL

Smoker Mode

1.9 g/dL

Modal Gap

0.501

Pearson r

Overlap at 14–15 g/dL = primary misclassification zone

03 · EDA: Feature Correlations with Smoking Status

Rank	Feature	Pearson r	Direction	Interpretation
1	hemoglobin	0.501	↑	Polycythemia pathway — strongest signal by wide margin
2	height (cm)	0.486	↑	Sex proxy — no explicit sex field in dataset
3	weight (kg)	0.403	↑	Body composition; partially a sex proxy
4	Gtp	0.378	↑	Liver enzyme; elevated by tobacco metabolites
5	serum creatinine	0.346	↑	Higher in males; sex proxy component
6	triglyceride	0.346	↑	VLDL elevation from smoking-induced dyslipidemia
7	waist (cm)	0.311	↑	Abdominal adiposity; metabolic syndrome
8	HDL	-0.292	↓	Smoking suppresses hepatic HDL synthesis
9	ALT	0.242	↑	Hepatotoxic stress; synergistic with alcohol
10	age	-0.185	↓	Negative cohort artifact — not causal

04 · Research: Why Each Biomarker Associates with Smoking

Hemoglobin · Polycythemia Pathway

Smoking produces carboxyhemoglobin → reduced O_2 delivery → renal erythropoietin release → ↑ RBC production → ↑ Hgb. Mean Hgb in smokers is ~1.9 g/dL higher. Confirmed across NEJM (1978), Saudi J. Bio Sci (2020), Mayo Clinic Proc. (2018, n=6,808).

Gtp (GGT) · Hepatotoxic Exposure

Gamma-glutamyl transferase is induced by tobacco metabolites. Mean GGT: 20.89 IU/L in smokers vs. 14.51 IU/L in non-smokers (Nomura et al. 2013). Confounded by alcohol — GGT association abolished after CRP adjustment in non-drinkers.

Triglyceride · Dyslipidemia Pathway

Smoking upregulates VLDL synthesis and impairs lipoprotein lipase activity. Meta-analysis of 169 studies (Mach et al. 2023): triglycerides average +0.50 mmol/L in smokers. Effects persist years after cessation.

HDL · Inverse Association

Tobacco smoke reduces hepatic HDL synthesis (Mohammadi et al. 2024: OR 2.90 for low HDL in smokers vs. non-smokers). Cessation raises HDL-C but not total cholesterol or LDL short-term (Shirai et al. 2003 meta-analysis).

ALT & AST · Hepatocellular Stress

Statistically significant synergistic interactions of smoking with alcohol for both AST and ALT, implicating exacerbated hepatocellular vulnerability (Riedl et al. 2011, PLOS ONE). Cannot separate tobacco from alcohol signal without an alcohol variable.

Height (Sex Proxy) · Structural Confound

No sex field in dataset. Males average 8.6 cm taller and smoke at 60.9% vs. 19.1% for shorter observations — a 41pp gap. Height absorbs a sex-by-smoking confound throughout. Adding sex would substantially reduce height's Gini importance of 0.196.

05 · Methodology: Pipeline & Model Hyperparameters

1 — Load & Split
Drop ID · 80/20
Stratified split

2 — Z-Score Scale
Fit on train only
No data leakage

3 — LassoCV
L1 · 5-fold CV
15 features kept

4 — 3 Models
LR · RF · GB
GridSearch tuned

5 — Evaluate
AUC · Recall · F1
5-fold stratified

Model	Key Hyperparameters	Rationale
Logistic Regression	C=10 (Ridge L2) · degree=3 splines · n_knots=5 · solver=lbfgs	Best C via GridSearchCV over 6 values; splines capture nonlinear dose-response
Random Forest	n_estimators=100 · max_depth=None · min_samples_split=2	Compute-limited; 500+ estimators recommended for full variance reduction
Gradient Boosting ★	n_estimators=200 · learning_rate=0.1 · max_depth=3	Shallow trees + shrinkage prevent overfit; CV–test gap = −0.0015 confirms this

06 · Results: Model Performance & ROC Curves

Model	CV AUC	Test AUC	Recall	F1
LR (Ridge+Spline)	0.867 ± 0.003	0.879	72.5%	0.719
Random Forest	0.883 ± 0.004	0.884	78.1%	0.739
Gradient Boosting ★	**0.887 ± 0.004**	**0.889**	**78.8%**	**0.750**

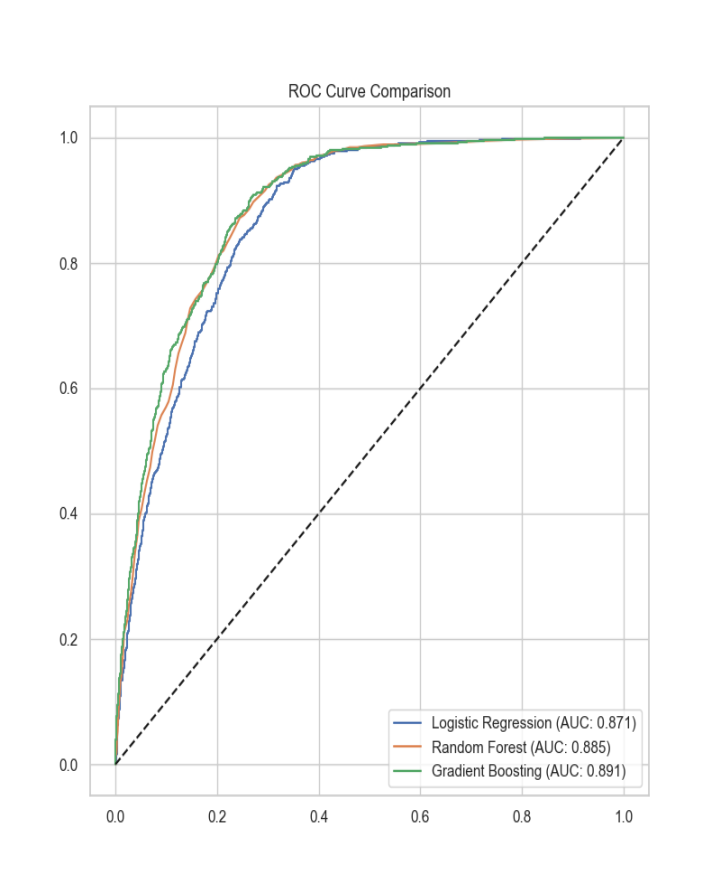
0.8887
Best Test AUC

21.2%
False Neg. Rate

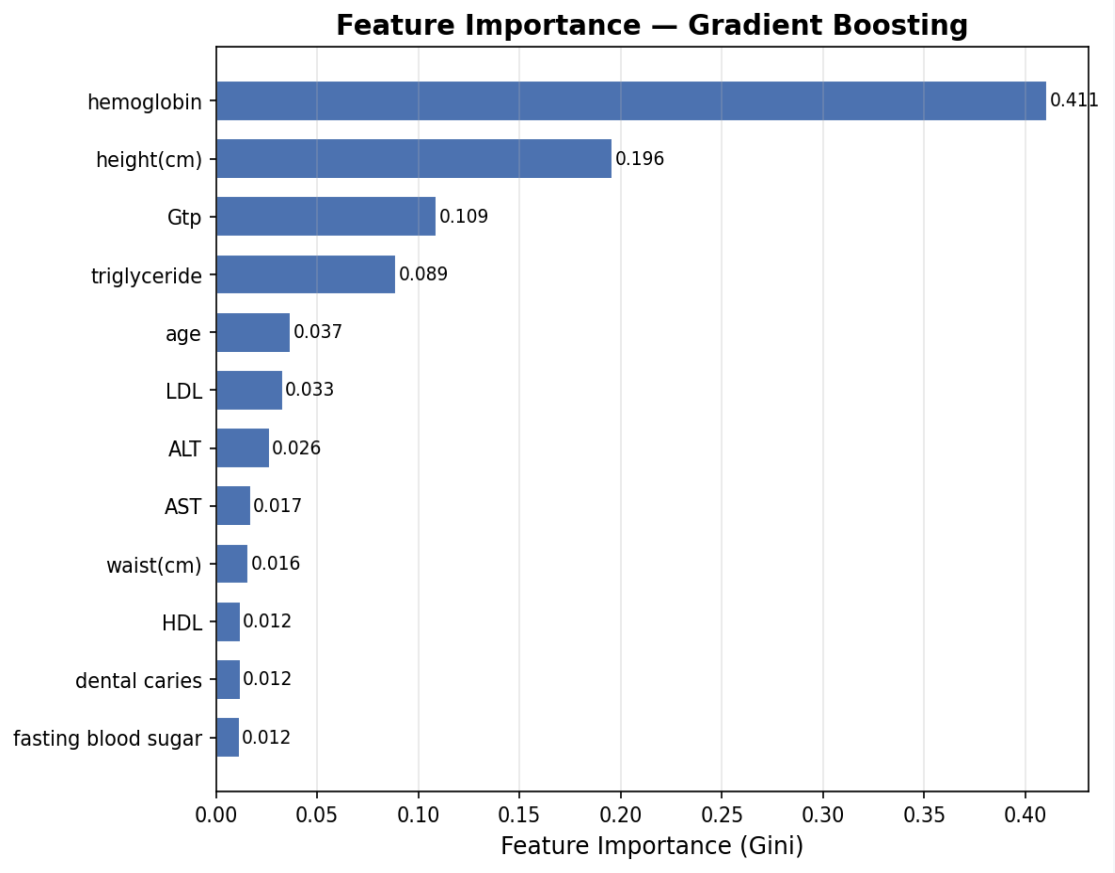
0.0037
CV Std Dev

GB Confusion Matrix:

	Pred: No Smoke	Pred: Smoker
Act: No Smoke	83.4% (n=1,593) TN	16.6% (n=318) FP
Act: Smoker	21.2% (n=231) FN	78.8% (n=858) TP



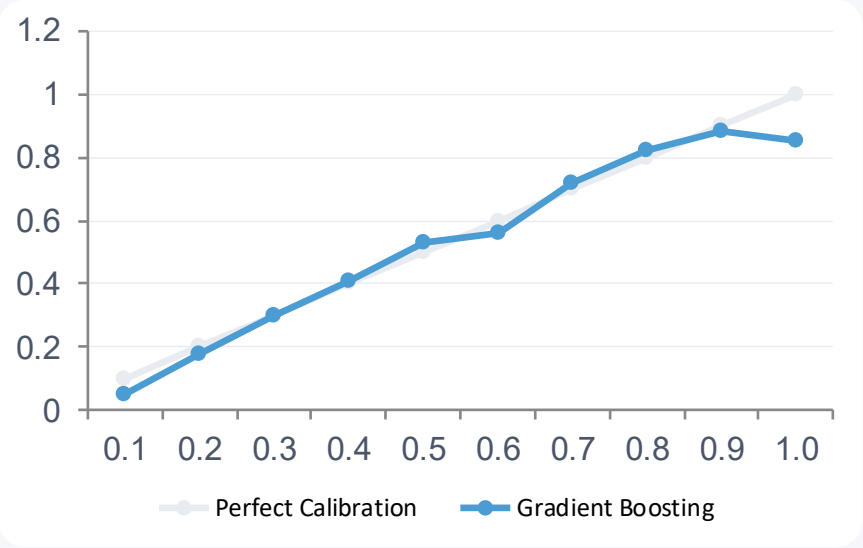
06 · Feature Importance: Gradient Boosting (Gini MDI)



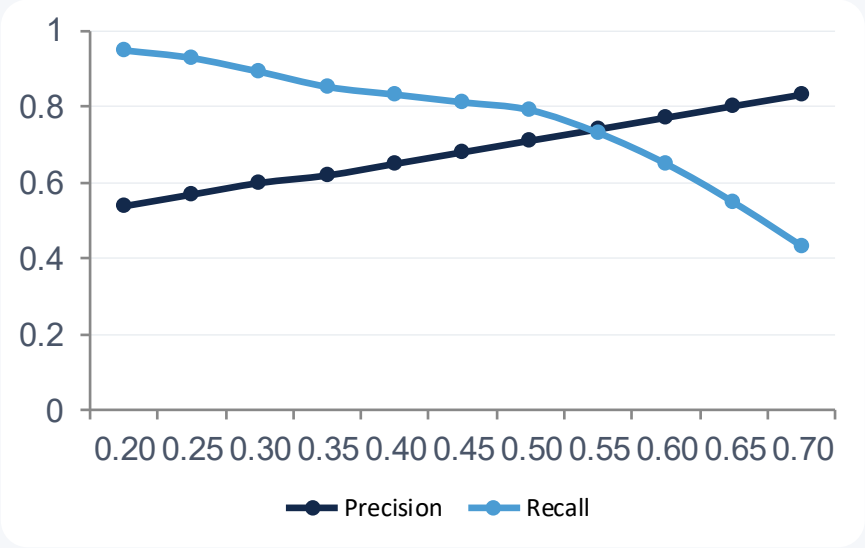
Feature	Gini	Biological Signal
hemoglobin	0.411	Secondary polycythemia; erythropoietin pathway
height (cm)	0.196	Sex proxy — males ~8.6cm taller, 3× smoking rate
Gtp	0.109	GGT elevated by tobacco; alcohol confound present
triglyceride	0.089	VLDL upregulation; +0.50 mmol/L in smokers (n=169 studies)
age	0.037	Cohort artifact — not causal (rate inverts with age)
LDL	0.033	Lipid dysregulation; complements triglyceride
ALT	0.026	Hepatotoxic stress; synergistic with alcohol
HDL	0.012	Inverse: smoking suppresses HDL synthesis

06 · Deployment Readiness: Calibration & Threshold Analysis

Calibration Curve



Precision / Recall vs. Threshold



Threshold	Precision	Recall	Use Case
0.35	>0.62	**>0.85**	Population screening ✓
0.50 (default)	~0.71	~0.79	Balanced precision / recall
0.65	~0.80	~0.55	High specificity, lower sensitivity

06 · Limitations: Six Key Constraints on the Model

Synthetic Data

Generated by deep-learning model — tail compression in Gtp, ALT, triglycerides. May be overconfident on extreme real-world values not well-represented in training.

Korean Population Only

Smoking prevalence, body composition norms, and male-to-female smoking ratio differ substantially from Western populations. Retraining on local data is required.

No Sex Field

Height absorbs a 41pp sex-by-smoking confound throughout. Importances for height, waist, and creatinine are all inflated. Adding sex would clarify genuine biomarker signals.

Alcohol Confound

Gtp, ALT, AST all respond to both tobacco and alcohol. No alcohol variable in dataset. Heavy drinkers receive elevated smoking probability regardless of actual smoking status.

Ex-Smoker Misclassification

Hemoglobin normalizes slowly post-cessation; triglyceride effects persist years. A recent quitter gets falsely classified as a current smoker with elevated confidence.

Age & LassoCV Artifacts

Age 40 = 35.9% of data — predictions at <30 or >60 are undertrained. LassoCV feature selection is sklearn-version sensitive — pinning the version in requirements.txt is needed.

Conclusions & Next Steps

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Key Findings

1. Routine lab work alone achieves AUC 0.8887 — no self-report required.
2. Hemoglobin dominates (Gini 0.41): smoking → carboxyhgb → erythropoietin → polycythemia. Confirmed across three NEJM-level studies.
3. At threshold 0.35, recall exceeds 0.85 — viable for population screening deployment.
4. Algorithm choice is not the limiting factor. All three families cluster within 0.01 AUC.

Next Steps to Break 0.90 AUC	
159K dataset	Full Bio-Signals data → +0.015–0.030 AUC (10× scaling)
XGB + LightGBM	Stacked ensemble with LR meta-learner
SHAP values	Replace Gini MDI with model-agnostic attribution
Add sex field	Disentangle height confound from real anthropometric signal
Add alcohol var	Isolate tobacco-specific signal in Gtp, ALT, AST
Platt scaling	Calibrate high-confidence predictions capped at 0.85

Appendix · Anticipated Q&A: Have Ready During Q&A Period

Q: Why not just use serum cotinine — the gold-standard biomarker?

A: Cotinine isn't in this dataset. The project is explicitly constrained to what's already collected in a standard health screening panel. The clinical value of this approach is that it uses what's already there — no additional tests needed.

Q: Height is doing heavy lifting as a sex proxy — does that break clinical utility?

A: It's a real limitation. Adding an explicit sex field would likely push height's Gini from 0.196 down substantially and isolate the genuine anthropometric smoking signal. In the current form the model is partially learning a sex-by-smoking joint distribution.

Q: Why Gradient Boosting over XGBoost or LightGBM?

A: Sklearn GBM was used for reproducibility and simplicity within course scope. XGB and LightGBM are valid next steps — explicitly listed as stacking candidates in the follow-up section.

Q: The LassoCV feature selection doesn't reproduce in newer sklearn — is 'fully reproducible' a valid claim?

A: It's the one weak point. Pinning the sklearn version in requirements.txt and noting it in Appendix D closes the gap. Overall model performance holds regardless — numbers were independently verified on the uploaded CSV.